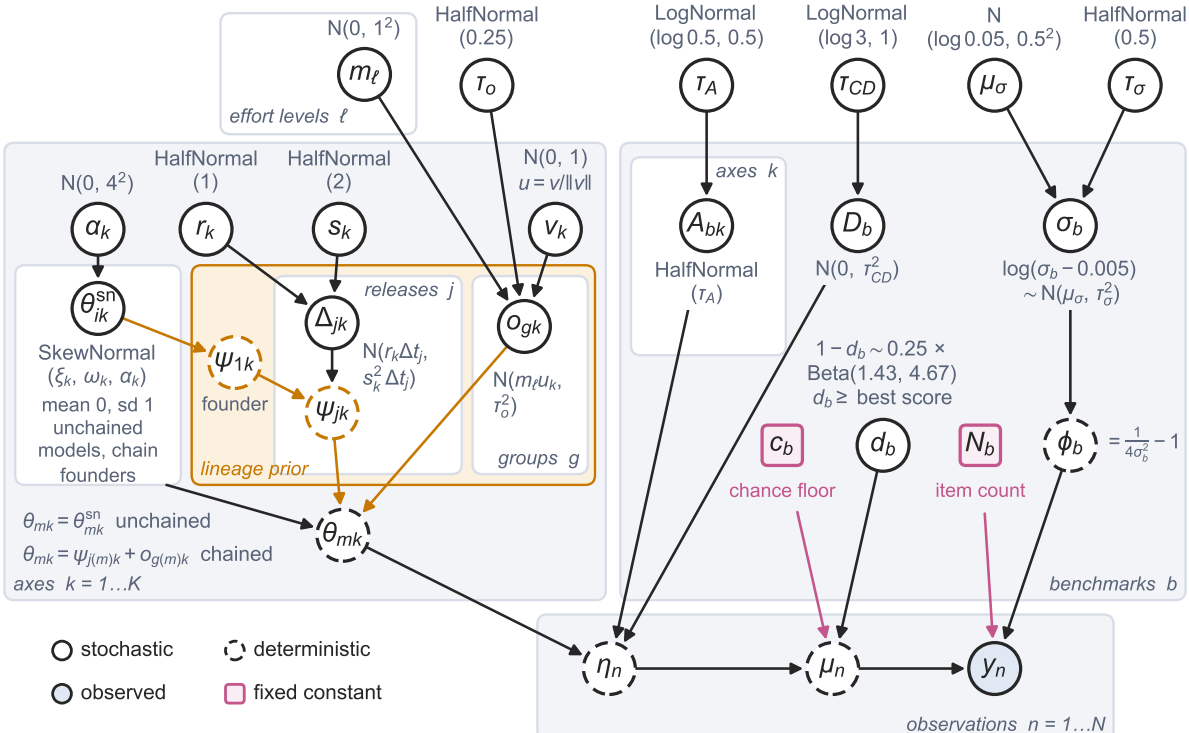

$$\eta_n = \sum_k A_{b_n k}\,\theta_{m_n k} - D_{b_n}$$

$$\mu_n = c_{b_n} + (d_{b_n} - c_{b_n})\,\sigma(\eta_n)$$

$$y_n \sim \mathrm{Beta}(\mu_n \phi_{b_n},\ (1-\mu_n)\,\phi_{b_n})$$ continuous score

$$N_{b_n} y_n \sim \mathrm{BetaBinomial}(N_{b_n},\ \mu_n \phi_{b_n},\ (1-\mu_n)\,\phi_{b_n})$$ score on N_b items